

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0799

Orchard Park 138 kV
November 02, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0799. The Interconnection Customer proposes a 132.0 MW battery energy storage facility interconnecting at the Orchard Park 138 kV substation in the Pennsylvania area. The request was received on May 17, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 316 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 38 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$62,014,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: REWEY / MILWAUKEE 138 KV L
Contingency: SUNRISE CENTER / WILLIM F MATSON GENERATING STA 500 KV
Loading: 171.0% (Rating: 218 MVA)
DFAX: 5.3%
Pre-project loading: 167.8%
2. Facility: PETERSON FLATS SUBSTATION / OTSEO 138 KV
Contingency: WEST WAYNESBORO / CHAMBERSBURG 345 KV
Loading: 156.8% (Rating: 253 MVA)
DFAX: 17.0%
Pre-project loading: 147.9%
3. Facility: ANTRIM / PETERSON FLATS 138 KV
Contingency: W WAYNESBORO / CHAMBERSBURG 345 KV
Loading: 155.7% (Rating: 264 MVA)
DFAX: 17.0%
Pre-project loading: 147.2%
4. Facility: CHAMBERSBURG 345/138 kV Transformer
Contingency: WEST WAYNESBORO / CHAMBERSBURG 345 KV
Loading: 154.4% (Rating: 503 MVA)
DFAX: 38.9%
Pre-project loading: 144.2%
5. Facility: HUTCHINSON PLANT 1 / YACHT CLUB D.S 138 KV
Contingency: WATFORD CITY / HUTCHINSON PLANT 1 138 KV Line
Loading: 148.7% (Rating: 282 MVA)
DFAX: 5.5%
Pre-project loading: 146.1%

6. Facility: CHAMBERSBURG / ORCHARD PARK 138 KV line
Contingency: N LEWISTON / CHAMBERSBURG 138 KV L
Loading: 145.9% (Rating: 181 MVA)
DFAX: 51.1%
Pre-project loading: 108.7%
7. Facility: CATOCTIN / ALTAVISTA POWER STA 138 KV line
Contingency: DRY GAP / W KELSO 138 KV
Loading: 144.9% (Rating: 260 MVA)
DFAX: 8.0%
Pre-project loading: 140.8%
8. Facility: YACHT CLUB D.S / ORRTANNA 138 KV
Contingency: HUTCHINSON PLANT 1 / WATFORD CITY 138 KV
Loading: 141.3% (Rating: 292 MVA)
DFAX: 5.5%
Pre-project loading: 138.8%
9. Facility: THURMONT / CATOCTIN 138 KV
Contingency: DRY AP / W KELSO 138 KV
Loading: 139.9% (Rating: 285 MVA)
DFAX: 8.0%
Pre-project loading: 136.2%
10. Facility: REWEY 345-138 kV XFMR
Contingency: CHAMBERSBURG 345-138 kV Transformer
Loading: 137.0% (Rating: 446 MVA)
DFAX: 5.0%
Pre-project loading: 135.5%
11. Facility: ESS WESEL / PINEVILLE 138 KV
Contingency: FAIR OAKS / CHAMBERSBURG 345 KV
Loading: 136.6% (Rating: 277 MVA)
DFAX: 5.4%
Pre-project loading: 134.0%
12. Facility: M105 / WYOMING 138 KV
Contingency: CHAMBERSBURG / W WAYNESBORO 345 KV
Loading: 134.5% (Rating: 394 MVA)
DFAX: 14.8%
Pre-project loading: 129.6%
13. Facility: THURMONT / DRY GAP 138 KV
Contingency: DRY GAP SUBSTATION / WEST KELSO 138 KV
Loading: 129.4% (Rating: 459 MVA)
DFAX: 14.1%
Pre-project loading: 125.4%
14. Facility: NORTH LEWISTON / CHAMBERSBURG 138 KV line
Contingency: CHAMBERSBURG / W WAYNESBORO 345 KV
Loading: 128.2% (Rating: 492 MVA)
DFAX: 23.3%
Pre-project loading: 121.9%
15. Facility: TEXAS EASTERN / ORCHARD PARK 138 KV
Contingency: CHAMBERSBURG / NORTH LEWISTON SUBSTATION 138 KV
Loading: 127.6% (Rating: 205 MVA)
DFAX: 48.9%
Pre-project loading: 96.1%
16. Facility: TEXAS EASTERN / ANTRIM 138 KV
Contingency: N LEWISTON / CHAMBERSBURG 138 KV

Loading: 127.5% (Rating: 193 MVA)
DFAX: 48.9%
Pre-project loading: 94.0%

17. Facility: WYOMING / N LEWISTON 138 KV

Contingency: CHAMBERSBURG / W WAYNESBORO 345 KV
Loading: 124.0% (Rating: 441 MVA)
DFAX: 14.8%
Pre-project loading: 119.6%

18. Facility: CARREL / LOURICH 138 KV

Contingency: FAIR OAKS / CHAMBERSBURG SUBSTATION 345 KV L
Loading: 123.6% (Rating: 376 MVA)
DFAX: 23.4%
Pre-project loading: 115.4%

19. Facility: FOUR HILLS / RUBIDOUX 138 KV

Contingency: W ELSON / DRY GAP SUBSTATION 138 KV Line
Loading: 120.6% (Rating: 540 MVA)
DFAX: 7.0%
Pre-project loading: 118.9%

20. Facility: MESSICK / FIVE CHANNELS 138 KV line

Contingency: OTSEGO / CARL BAILEY 138 KV
Loading: 119.9% (Rating: 363 MVA)
DFAX: 8.3%
Pre-project loading: 116.9%

21. Facility: ANTRIM / MTN VIEW 138 KV Line

Contingency: FAIR OAKS / CHAMBERSBURG SUBSTATION 345 KV
Loading: 119.2% (Rating: 434 MVA)
DFAX: 23.4%
Pre-project loading: 112.1%

22. Facility: RUBIDOUX / ALTAVISTA POWER STATION 138 KV

Contingency: WEST KELSON SUBSTATION / DRY GAP 138 KV
Loading: 119.0% (Rating: 568 MVA)
DFAX: 7.0%
Pre-project loading: 117.4%

23. Facility: MTN VIEW / LOURICH 138 KV

Contingency: CHAMBERSBURG / FAIR OAKS 345 KV
Loading: 118.0% (Rating: 447 MVA)
DFAX: 23.4%
Pre-project loading: 111.1%

24. Facility: CRESSEY / ESS WESEL 138 KV

Contingency: FAIR OAKS / CHAMBERSBURG 345 KV
Loading: 117.9% (Rating: 351 MVA)
DFAX: 5.4%
Pre-project loading: 115.8%

25. Facility: MESSICK / PLUESS 138 KV

Contingency: CARL BAILEY / OTSEGO 138 KV
Loading: 117.3% (Rating: 394 MVA)
DFAX: 8.3%
Pre-project loading: 114.6%

26. Facility: ATHALMER TAP / CAMERON 138 KV

Contingency: CARL BAILEY / OTSEGO 138 KV L
Loading: 114.9% (Rating: 430 MVA)

DFAX: 8.3%
Pre-project loading: 112.3%

27. Facility: DUNDEE / METRO CTR 138 KV

Contingency: CAMERON SUBSTATION / ATHALMER TAP 138 KV
Loading: 112.5% (Rating: 319 MVA)
DFAX: 5.7%
Pre-project loading: 110.2%

28. Facility: OTSEGO / M105 138 KV Line

Contingency: CHAMBERSBURG / WEST WAYNESBORO 345 KV line
Loading: 110.6% (Rating: 471 MVA)
DFAX: 14.8%
Pre-project loading: 106.4%

29. Facility: EL SOBRANTE / HOLLISTER 138 KV

Contingency: PINEVILLE / E HAGERSTOWN 138 KV
Loading: 110.0% (Rating: 411 MVA)
DFAX: 6.4%
Pre-project loading: 108.0%

30. Facility: E HAGERSTOWN / ATHALMER TAP 138 KV

Contingency: INNERARITY / PINEVILLE 138 KV
Loading: 110.0% (Rating: 532 MVA)
DFAX: 8.2%
Pre-project loading: 108.0%

31. Facility: GRAND PT / NORTH LEWISTON 138 KV

Contingency: WILLIAM F MATSON GENERATING STA / REWEY 345 KV
Loading: 109.7% (Rating: 182 MVA)
DFAX: 8.0%
Pre-project loading: 103.8%

32. Facility: ANTIETAM SUBSTATION / PINEVILLE 138 KV Line

Contingency: W WAYNESBORO / PINEVILLE 345 KV
Loading: 109.0% (Rating: 364 MVA)
DFAX: 8.8%
Pre-project loading: 105.8%

33. Facility: FAI OAKS / CHAMBERSBURG 345 KV

Contingency: CARREL 345/138 kV TX
Loading: 106.8% (Rating: 1,662 MVA)
DFAX: 21.4%
Pre-project loading: 105.1%

34. Facility: CARREL 345/138 kV Transformer Bank

Contingency: CHAMBERSBURG / FAIR OAKS 345 KV
Loading: 106.1% (Rating: 1,121 MVA)
DFAX: 16.6%
Pre-project loading: 104.2%

35. Facility: YOUNGWOOD 500-345 kV Transformer

Contingency: YOUNGWOOD / UNIV OF NC CHAPEL HILL COGEN FACILITY 345 KV Line
Loading: 105.5% (Rating: 732 MVA)
DFAX: 5.5%
Pre-project loading: 104.5%

36. Facility: HAMPTON RD / MEDOA 345 KV

Contingency: SIMPSON STUART TAP / FISHKILL PLAINS SUBSTATION 345 KV
Loading: 103.6% (Rating: 1,928 MVA)
DFAX: 7.1%

Pre-project loading: 103.1%

37. Facility: BEAVER / EUFAULA 500 KV Line
Contingency: POST ROCK / NEWPORT TIE 230 KV
Loading: 100.6% (Rating: 3,064 MVA)
DFAX: 8.6%
Pre-project loading: 100.3%

38. Facility: OTSGO / EAST WAYNESBORO 138 KV
Contingency: CHAMBERSBURG 345-138 kV Auto
Loading: 100.0% (Rating: 601 MVA)
DFAX: 33.0%
Pre-project loading: 92.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
REWEY / MILWAUKEE 138 KV L	Rebuild line with higher-capacity conductor	\$5,699,000	43% (\$2,442,000)
OTSEGO / PETERSON FLATS 138 KV Line	Rebuild line with higher-capacity conductor	\$6,813,000	39% (\$2,658,000)
ANTRIM / PETERSON FLATS 138 KV	Rebuild line with higher-capacity conductor	\$7,013,000	39% (\$2,736,000)
CHAMBERSBURG 345 KV/138 kV Auto	Replace transformer with higher-rated unit	\$14,475,000	19% (\$2,715,000)
HUTCHINSON PLANT 1 / YACHT CLUB D.S 138 KV	Rebuild line with higher-capacity conductor	\$19,438,000	34% (\$6,677,000)
CHAMBERSBURG / ORCHARD PARK 138 KV line	Rebuild line with higher-capacity conductor	\$4,832,000	85% (\$4,102,000)
CATOCTIN / ALTAVISTA POWER STATION 138 KV L	Rebuild line with higher-capacity conductor	\$6,386,000	41% (\$2,635,000)
YACHT CLUB D.S / ORRTANNA 138 KV	Rebuild line with higher-capacity conductor	\$1,896,000	34% (\$651,000)
THURMONT / CATOCTIN 138 KV	Rebuild line with higher-capacity conductor	\$4,661,000	41% (\$1,923,000)
REWEY 345-138 kV XFMR	Replace transformer with higher-rated unit	\$26,662,000	45% (\$12,076,000)
ESS WESEL / PINEVILLE 138 KV	Rebuild line with higher-capacity conductor	\$3,463,000	4% (\$129,000)
M105 / WYOMING 138 KV	Rebuild line with higher-capacity conductor	\$1,267,000	13% (\$159,000)
THURMONT / DRY GAP 138 KV	Rebuild line with higher-capacity conductor	\$3,671,000	12% (\$433,000)
N LEWISTON / CHAMBERSBURG SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$4,408,000	16% (\$683,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ORCHARD PARK / TEXAS EASTERN SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$1,975,000	84% (\$1,665,000)
ANTRIM / TEXAS EASTERN 138 KV Line	Rebuild line with higher-capacity conductor	\$8,662,000	84% (\$7,302,000)
WYOMING / N LWISTON 138 KV	Rebuild line with higher-capacity conductor	\$1,205,000	13% (\$151,000)
CARREL / LOURICH 138 KV	Rebuild line with higher-capacity conductor	\$7,100,000	12% (\$818,000)
FOUR HILLS / RUBIDOUX 138 KV	Rebuild line with higher-capacity conductor	\$5,065,000	24% (\$1,215,000)
MESSICK / FIVE CHANNELS SUBSTATION 138 KV Line	Rebuild line with higher-capacity conductor	\$2,859,000	7% (\$190,000)
MTN VIEW / ANTRIM 138 KV	Rebuild line with higher-capacity conductor	\$7,123,000	12% (\$866,000)
RUBIDOUX / ALTAVISTA POWER STATION 138 KV	Rebuild line with higher-capacity conductor	\$3,271,000	24% (\$784,000)
MTN VIEW / LOURICH 138 KV	Rebuild line with higher-capacity conductor	\$5,585,000	12% (\$679,000)
CRESSEY / ESS WESEL 138 KV	Rebuild line with higher-capacity conductor	\$8,474,000	4% (\$316,000)
MESSICK / PLUESS 138 KV	Rebuild line with higher-capacity conductor	\$3,876,000	7% (\$257,000)
ATHALMER TAP / CAMERON 138 KV	Rebuild line with higher-capacity conductor	\$4,212,000	7% (\$280,000)
DUNDEE / METRO CTR 138 KV	Rebuild line with higher-capacity conductor	\$26,865,000	6% (\$1,518,000)
OTSEGO / M105 138 KV	Rebuild line with higher-capacity conductor	\$9,472,000	13% (\$1,192,000)
EL SOBRANTE / HOLLISTER 138 KV	Rebuild line with higher-capacity conductor	\$8,247,000	6% (\$470,000)
ATHALMER TAP / E HAGERSTOWN 138 KV	Rebuild line with higher-capacity conductor	\$7,553,000	5% (\$403,000)
GRAND PT / NORTH LEWISTON 138 KV	Reconductor line	\$722,000	100% (\$722,000)
ANTIETAM / PINEVILLE 138 KV	Reconductor line	\$1,560,000	7% (\$105,000)
CHAMBESBURG / FAIR OAKS 345 KV L	Reconductor line	\$14,711,000	3% (\$400,000)
CARREL 345/138 kV Transformer Bank	Replace transformer with higher-rated unit	\$11,731,000	4% (\$414,000)
YOUNGWOOD 500-345 kV Transformer	Replace transformer with higher-rated unit	\$31,988,000	5% (\$1,528,000)
HAMPTON RD / MEDORA SUBSTATION 345 KV	Reconductor line	\$13,045,000	1% (\$159,000)
BEAVER SUBSTATION / EUFAULA 500 KV	Reconductor line	\$20,259,000	1% (\$209,000)
OTSGO / EAST WAYNESBORO 138 KV	Reconductor line	\$1,700,000	21% (\$352,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$62,014,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.